

Evaluating the Memory/Latency Boundary of Adaptive Symbol-Table Name Representations in Embedded Workloads

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Abstract—Symbol tables are a core compiler data structure whose physical memory footprint and lookup latency directly govern execution efficiency under severe resource constraints. Modern 64-bit systems rely on `std::unordered_map` with Short String Optimization (SSO), storing small identifiers (≤ 15 bytes) without secondary heap allocations. In this paper, we investigate the fundamental memory and latency trade-offs of adaptive symbol-table representations in embedded compiler and runtime environments. We formulate the primary research question: *Under what workload conditions can adaptive symbol-table name representations overcome the structural memory efficiency of an embedded conventional hash-table baseline while satisfying a latency constraint?*

We present **SymTabV3**, an adaptive scope-aware symbol table that dynamically assigns symbols to `INLINE`, `INTERNEDED`, or front-coded `COMPRESSED` representations with zero-allocation stack buffer decoding. We derive a closed-form analytical break-even model: for identifiers longer than the SSO threshold ($L > 15$ B), adaptive storage beats Conventional hash tables when the live duplication ratio satisfies $k > k_{\text{breakeven}} = (2L + 86)/(L + 13)$. We evaluate our model using physical allocator heap tracking (`malloc_usable_size`) across four real embedded workloads (FREERTOS, ARDUINO, ZEPHYR, ESP-IDF), 20 host software corpora, and controlled synthetic experiments (Workloads A–F).

On the large ZEPHYR workload (2.6M events, 228,739 unique symbols), **SymTabV3** reduced physically measured final heap by 20.2% (13.52 MB) relative to the embedded conventional baseline (53.39 MB vs 66.91 MB final heap; 57.89 MB vs 76.37 MB peak heap). This reduction was achieved with sub-microsecond median lookup latency ($p_{50} = 0.066 \mu\text{s}$), while the p_{95} latency increased from $0.125 \mu\text{s}$ to $0.228 \mu\text{s}$ ($1.824\times$). Thus, the architecture demonstrates a measurable RAM advantage in the large-scale embedded regime, but does not satisfy the $1.25\times p_{95}$ latency constraint due to measured reconstruction/copy overhead. Additionally, we evaluate **SymTabV4** as a disclosed negative result, showing that side-table container overheads (T_{table}) erode core scalar field savings across real identifier mixes.

Index Terms—Symbol Table, Compiler Memory Management, Embedded Systems, Short String Optimization, String Interning, Front Coding, Data Structure Break-Even Boundary, Empirical Evaluation.

I. INTRODUCTION

Compilers and language interpreters repeatedly create, resolve, and reclaim symbolic bindings during lexical, syntactic, and semantic analysis [1]. A symbol table maps identifier names to binding metadata across nested lexical scopes. In resource-constrained embedded systems—such as microcon-

troller toolchains, real-time operating systems (RTOS), and on-device interpreters—available RAM is severely limited.

In desktop and server host compilers (e.g., LLVM [2]), standard C++ string implementations rely on Short String Optimization (SSO) [3], [4]. Identifiers up to 15 bytes are stored directly inside the 24-byte string structure, avoiding heap allocations. Consequently, conventional hash maps (`std::unordered_map<std::string, SymbolMeta>`) pay zero secondary heap string allocation cost for short names.

In embedded environments, however, secondary heap overheads and container metadata can consume a significant fraction of total memory. While string interning [5], [6] and front-coded block compression [7], [8] reduce string redundancy, their physical RAM savings and tail-latency costs under realistic scope lifecycles have lacked rigorous physical allocator measurement.

In this work, we formulate and address the following central research question:

Under what workload conditions can adaptive symbol-table name representations overcome the structural memory efficiency of an embedded conventional hash-table baseline while satisfying a latency constraint?

We make the following main contributions:

- 1) **Adaptive Architecture (SymTabV3)**: We present a C++ symbol table architecture featuring scope-lifetime slot recycling, 3-tier representation routing (`INLINE`, `INTERNEDED`, `COMPRESSED`), 1-byte hash fingerprints, and zero-allocation stack buffer decoding.
- 2) **Analytical Cost & Break-Even Model**: We derive closed-form memory equations proving that for $L \leq 15$ B no positive break-even live duplication ratio k exists, whereas for $L > 15$ B, interning breaks even when $k > (2L + 86)/(L + 13)$.
- 3) **Embedded Memory Advantage**: On the large ZEPHYR RTOS workload, **SymTabV3** achieves a 20.2% (13.52 MB) physical final heap reduction (and 24.2% peak heap reduction) over an optimized embedded conventional baseline.

- 4) **Quantified Memory/Latency Trade-off:** We demonstrate sub-microsecond median lookups ($p_{50} = 0.066 \mu s$) on ZEPHYR, while reporting that p_{95} tail latency increases to $0.228 \mu s$ (vs $0.125 \mu s$ for the embedded conventional baseline), failing the $1.25 \times p_{95}$ gate due to measured reconstruction/copy overhead.
- 5) **Disclosed Negative Architectural Result (SymTabV4):** We present SymTabV4 (representation-conditional metadata) as a disclosed negative result, showing that sparse side-table allocation floors (T_{table}) outweigh core struct savings across real representation mixes.
- 6) **Controlled Synthetic & 20-Corpus Host Validation:** We confirm theoretical predictions via synthetic microbenchmarks A–F ($k \geq 3.33$ at $L = 32 \text{ B}$) and 20 host software corpora.

II. BACKGROUND AND MOTIVATION

A. Conventional Symbol Tables and SSO

A conventional symbol table uses a hash map from string keys to symbol attributes. In 64-bit systems, a `std::string` instance occupies 32 bytes (or 24 bytes under `libc++`), consisting of a pointer, length, and capacity/inline buffer. When identifier length $L \leq 15$ bytes, characters reside inside the inline buffer (SSO). For $L > 15$ bytes, `std::string` allocates a heap buffer of size $L+1$ bytes plus allocator header overhead $H \approx 16 \text{ B}$.

In host standard libraries, `std::unordered_map` node allocations incur node overhead ($S_{\text{node}} \approx 32 \text{ B}$). For short names ($L \leq 15 \text{ B}$), Conventional pays only metadata and node footprint:

$$M_{\text{conv_host}}(k, L) \approx k \cdot (S_{\text{meta}} + S_{\text{node}}) \quad \text{for } L \leq 15 \text{ B} \quad (1)$$

For embedded targets, we construct `EmbeddedConventional`, a compact baseline combining 16-byte fixed entries, a contiguous byte arena, and an open-addressing slot index.

B. String Interning

String interning maintains a global pool of unique string instances. Each symbol record stores a 4-byte pool index. However, interning incurs two fixed costs: 1) a global pool vector; 2) a lookup map (`poolLookup_`) paying node overhead $S_{\text{pool_node}} \approx 68 \text{ B}$ per unique name.

The memory for k live occurrences of a unique name of length L under interning is:

$$C_{\text{pool}}(L) \approx 2L + 86 \quad (2)$$

$$M_{\text{int}}(k, L) \approx k \cdot C_{\text{idx}} + C_{\text{pool}}(L) \quad (3)$$

where $C_{\text{idx}} \approx 28 \text{ B}$ is the index slot overhead.

III. RELATED WORK

A. Compiler Symbol Tables

Scope management and binding resolution have been studied since early structured language compilers [1], [13], [14]. Traditional scope stacks push a hash map on scope entry and pop it on exit. LLVM’s `IdentifierTable` [2] utilizes a global string map backed by a bump-pointer allocator (`BumpPtrAllocator`), prioritizing allocation throughput but allowing memory to grow monotonically until compilation completes.

B. String Compression and Dictionary Search

String interning is standard in runtimes such as Java HotSpot [5] and CPython [6]. For text indexing, front-coded block compression [7]–[9] stores lexicographically shared prefixes relative to periodic anchor strings. Succinct data structures [10], [11] and minimal perfect hash functions (e.g., CHD [12]) achieve optimal static byte density but cannot handle dynamic lexical scope entry and exit.

C. Cache-Conscious Hash Tables

Open-addressing structures such as Robin Hood hashing [15], Cuckoo hashing [16], HAT-trie [17], and burst tries [18] optimize L1/L2 cache locality. SymTabV3 complements these by integrating scope-lifetime slot recycling, 3-tier representation routing, 1-byte hash fingerprints, and zero-allocation stack buffer decoding.

IV. ADAPTIVE SYMBOL-TABLE ARCHITECTURE

A. SymTabV3 Core Design

SymTabV3 organizes entries into a scope-managed slot arena. Each slot pays a 32-byte `PackedEntry` containing scope ID, access counters, representation tags, and a 12-byte payload union:

- **INLINE:** Identifiers ≤ 12 bytes are stored inside the slot union (`inlineBytes[12]`), requiring zero secondary allocations.
- **INTERNED:** Duplicate or hot symbols store a 4-byte index into a global pool.
- **COMPRESSED:** Long, cold symbols are front-coded in blocks of size $B = 32$ with anchor interval $A = 8$.

Scope exit events (`exitScope()`) recycle slot indices immediately into a LIFO free-list, preventing fragmentation. To avoid unnecessary decompression during lookup, SymTabV3 checks a 1-byte hash fingerprint array (`fp8`), rejecting non-matching candidates in $O(1)$ time.

B. Zero-Allocation Compressed Path Optimization

To minimize front-coded decode latency, SymTabV3 implements zero-allocation stack buffer decoding (`decodeToBuffer`). Rather than instantiating temporary `std::string` objects or executing substring concatenations (`cur.substr(0, len) + suffix`), prefix and suffix segments are copied directly into a stack-allocated buffer (`char stackBuf[512]`) via fast `memcpy`. Furthermore,

anchor indices are computed in $O(1)$ time (start = slot – (slot (mod A))), eliminating backward scanning loops. Name equality (`nameEquals`) performs direct in-place byte comparisons against `stackBuf`.

C. SymTabV4: Disclosed Negative Result

To test whether removing hot/cold metadata from `INLINE` entries improves byte density, we designed `SymTabV4`. `SymTabV4` reduces `PackedEntry` from 32 B to 24 B (`CoreEntry`) by moving scalar metadata (`accessCount`, `lastAccessEpoch`) to a sparse side table (`HotMetaTable`). Both node-based and Robin Hood open-addressing side-table variants were implemented and differentially verified against `SymTabV3` under `ASan/UBSan` across 200,000 trace steps.

V. ANALYTICAL COST AND BREAK-EVEN MODEL

Equating $M_{\text{int}}(k, L) < M_{\text{conv}}(k, L)$ yields the theoretical break-even condition:

$$k > k_{\text{breakeven}}(L) = \frac{C_{\text{pool}}(L)}{S_{\text{meta}} + S_{\text{str}}(L) - C_{\text{idc}}} \quad (4)$$

For short names ($L \leq 15$ B), Conventional string storage $S_{\text{str}}(L) = 0$ due to SSO. Substituting $S_{\text{str}}(L) = 0$ yields:

$$k > \frac{2L + 86}{24 - 28} = \frac{2L + 86}{-4} < 0 \quad (5)$$

Because the denominator is negative, **no positive live duplication ratio k exists for $L \leq 15$ B**. Interning cannot outperform Conventional hash tables on short SSO identifiers.

For long names ($L > 15$ B), Conventional pays heap allocation $S_{\text{str}}(L) = L + 1 + H \approx L + 17$. The break-even equation becomes:

$$k_{\text{breakeven}}(L) = \frac{2L + 86}{24 + (L + 17) - 28} = \frac{2L + 86}{L + 13} \quad (6)$$

As $L \rightarrow 15^+$, $k_{\text{breakeven}} \rightarrow 4.14$. As $L \rightarrow \infty$, $k_{\text{breakeven}} \rightarrow 2.0$. For $L = 32$ B, $k_{\text{breakeven}} = 150/45 \approx 3.33$.

For `SymTabV4` to beat `SymTabV3`, core-struct savings across `INLINE` entries must exceed the side table’s allocation floor (T_{table}) and entry cost ($S_{\text{hotmeta_marginal}} \approx 34\text{--}57$ B):

$$k_{\text{inline}} \cdot 8 \text{ B} > k_{\text{hot}} \cdot S_{\text{hotmeta_marginal}} + T_{\text{table}}(N_{\text{hot}}) \quad (7)$$

Because real software codebases maintain 10%–70% non-`INLINE` symbols, k_{hot} is non-zero, causing side-table overheads to exceed core scalar field savings.

VI. EXPERIMENTAL METHODOLOGY

A. Physical Allocator Heap Accounting

All heap measurements use a physical allocator interceptor (`include/heap_counter.hpp`) overriding global operator `new` and operator `delete`. The interceptor queries `malloc_usable_size(p)` to record true physical heap allocations, accounting for system page alignment and chunk rounding.

B. Workload Suite

We evaluate three distinct workload categories:

- 1) **Real Embedded Workloads:** Four real-world embedded software event traces: `FREERTOS` (227k events, 10,386 unique symbols), `ARDUINO` (94k events, 11,000 unique symbols), `ZEPHYR` (2.6M events, 228,739 unique symbols), and `ESP-IDF` (2.2M events, 231,075 unique symbols).
- 2) **Host Software Corpora Characterization:** 20 real software codebases evaluated on 64-bit host systems to assess identifier length distributions and SSO fractions.
- 3) **Synthetic Microbenchmarks (Workloads A–F):** Controlled experiments varying identifier length $L \in \{4 \dots 64\}$ B, duplication ratio $k \in \{1 \dots 20\}$, scope depth, and access skew ($N = 30$ seeds).

Throughout this paper, we explicitly distinguish **peak heap** (maximum RAM allocated during execution) from **final heap** (residual RAM after processing complete event traces).

VII. EMPIRICAL RESULTS

A. Primary Embedded Evaluation

Table I presents physical allocator heap measurements, lookup latencies, representation counts, and gate status across the four embedded workloads.

B. Physical Memory Victory on Zephyr

On the large `ZEPHYR` workload (2.6M events, 228,739 unique symbols), `SymTabV3` achieves a clear physical allocator memory victory. `SymTabV3` reduces final heap consumption by ****20.2% (13.52 MB)**** relative to `EmbeddedConventional` (53.39 MB vs 66.91 MB final heap) and by ****34.3% (27.83 MB)**** relative to `Interned` (53.39 MB vs 81.22 MB final heap). Peak heap memory is similarly reduced by 24.2% (57.89 MB vs 76.37 MB).

Representation routing on `ZEPHYR` assigns 213,614 symbols (93.4%) to `INLINE`, 5,372 (2.3%) to `INTERNEED`, and 9,753 (4.3%) to front-coded `COMPRESSED`.

C. Quantified Latency Trade-off and Reconstruction Overhead

The zero-allocation compressed-path optimization reduced `ZEPHYR` p_{95} latency from 0.547 μs down to 0.228 μs without altering memory footprint. Median lookup latency remains sub-microsecond ($p_{50} = 0.066 \mu\text{s}$).

However, comparing `SymTabV3` p_{95} (0.228 μs) against `EmbeddedConventional` p_{95} (0.125 μs) yields a ratio of ****1.824 $\times$ ****, failing the $1.25 \times p_{95}$ latency constraint.

This latency gap is driven by measured reconstruction/copy overhead. During execution, 8.81% of total lookup operations hit `COMPRESSED` entries. Reconstructing front-coded members requires copying multiple prefix and suffix segments across block members (mean depth 3.31 steps to anchor, $p_{95} = 7$ steps). Even with zero dynamic memory allocation, copying multiple memory slices introduces measured reconstruction/copy overhead that prevents satisfying the $1.25 \times$ gate.

TABLE I
CANONICAL PHYSICAL ALLOCATOR-MEASURED MEMORY AND LOOKUP LATENCIES ACROSS FOUR EMBEDDED WORKLOADS.

Workload	Implementation	Unique Syms	Peak Heap	Final Heap	p_{50} (μ s)	p_{95} (μ s)	p_{99} (μ s)	Mean (μ s)	Final RAM vs EmbConv	p_{95} Gate
FreeRTOS	ConventionalHost	10,386	3.99 MB	3.74 MB	0.072	0.189	0.295	0.089	–	–
	EmbeddedConventional	10,386	4.23 MB	3.89 MB	0.051	0.094	0.149	0.057	Baseline	Baseline
	Interned	10,386	5.17 MB	5.09 MB	0.072	0.143	0.203	0.087	+30.8%	FAIL
	SymTabV3 (Optimized)	10,386	4.78 MB	4.72 MB	0.071	0.277	0.445	0.101	+21.2%	FAIL
Arduino	ConventionalHost	11,000	2.42 MB	2.42 MB	0.080	0.192	0.276	0.093	–	–
	EmbeddedConventional	11,000	2.31 MB	2.27 MB	0.046	0.089	0.130	0.051	Baseline	Baseline
	Interned	11,000	3.70 MB	3.70 MB	0.070	0.135	0.195	0.079	+63.0%	FAIL
	SymTabV3 (Optimized)	11,000	2.99 MB	2.99 MB	0.061	0.171	0.356	0.077	+31.7%	FAIL
Zephyr	ConventionalHost	228,739	50.92 MB	47.76 MB	0.096	0.339	0.680	0.133	–	–
	EmbeddedConventional	228,739	76.37 MB	66.91 MB	0.050	0.125	0.225	0.060	Baseline	Baseline
	Interned	228,739	81.52 MB	81.22 MB	0.082	0.233	0.499	0.106	+21.4%	FAIL
	SymTabV3 (Optimized)	228,739	57.89 MB	53.39 MB	0.066	0.228	0.481	0.092	-20.2%	FAIL
ESP-IDF	ConventionalHost	231,075	73.28 MB	65.65 MB	0.103	0.255	0.536	0.123	–	–
	EmbeddedConventional	231,075	75.77 MB	67.82 MB	0.061	0.236	0.374	0.086	Baseline	Baseline
	Interned	231,075	100.21 MB	92.95 MB	0.090	0.216	0.573	0.113	+37.1%	FAIL
	SymTabV3 (Optimized)	231,075	89.36 MB	81.91 MB	0.079	0.297	0.712	0.113	+20.8%	FAIL

Quantitatively, SymTabV3 trades 0.103μ s of added p_{95} latency for 13.52 MB of physical final heap saved (131.26 MB saved per μ s of added p_{95} latency).

D. Disclosed Negative Result (SymTabV4)

SymTabV4 (representation-conditional side-table metadata) failed to beat SymTabV3 on 19/20 real software corpora and lost to Conventional on 20/20. Sparse side-table container allocations (T_{table}) plus marginal entry costs ($\sim 34\text{--}57$ B) exceed the 8 B core-struct savings across real identifier mixes.

E. Controlled Synthetic Microbenchmarks (Workloads A–F)

Synthetic experiments D1 and D2 confirm our analytical model. For $L = 32$ B, interning and SymTabV3 beat Conventional on peak heap at $k \geq 3$ (1.09 MB vs 1.16 MB) and on final heap at $k \geq 4$ (0.70 MB vs 1.44 MB), validating the theoretical break-even threshold $k_{\text{breakeven}} = 3.33$.

F. Supporting Host-Side 20-Corpus Characterization

On 64-bit host systems, ConventionalHost (`std::unordered_map`) achieves the lowest physical heap footprint on 20/20 real software corpora. Real host codebases are heavily dominated by short SSO identifiers (59%–76%) and low live duplication ($k < 1.5$), placing them strictly inside Conventional’s optimal memory regime.

VIII. DISCUSSION AND WORKLOAD BOUNDARIES

Our findings establish clear application boundaries for symbol table architectures:

- 1) **Host Compilers & Short Workloads:** When short SSO identifiers dominate ($L \leq 15$ B) and live duplication is low ($k < 1.5$), Conventional hash tables are optimal.
- 2) **Large-Scale Embedded Workloads:** When scale and identifier lengths are large ($L > 15$ B, $N > 200$ K), adaptive front-coding (SymTabV3) achieves significant physical RAM savings (20.2% / 13.52 MB on ZEPHYR).

IX. LIMITATIONS

1) Front-coded block decompression introduces measured reconstruction/copy overhead during tail lookups; 2) Global pool strings persist for compilation unit lifecycles; 3) 64-bit hash fingerprints carry a theoretical collision probability of $\approx 4 \times 10^{-10}$ at 260,000 symbols.

X. CONCLUSION

On the large ZEPHYR workload, SymTabV3 reduced physically measured final heap by 20.2% (13.52 MB) relative to the embedded conventional baseline. This reduction was achieved with sub-microsecond median lookup latency, while the p_{95} latency increased from 0.125μ s to 0.228μ s. Thus the architecture demonstrates a measurable RAM advantage in the large-scale embedded regime, but does not satisfy the $1.25 \times p_{95}$ latency constraint.

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